

Far-field boundary conditions in the Cerisse compressible solver: principle, implementation, and a one-dimensional reflection benchmark

Cerisse IBM validation campaign — BC sub-study

June 2026

Abstract

The Cerisse Cartesian-AMR immersed-boundary solver offers three usable far-field ghost-cell boundary conditions: zero-gradient extrapolation (**foextrap**), hard Dirichlet freestream, and an algebraic Riemann-invariant closure labelled “NSCBC”. This note (i) states the mathematical principle and exact **inputs** usage of each, (ii) benchmarks their acoustic reflection with the canonical Poinot–Lele one-dimensional pulse test, and (iii) identifies why the algebraic “NSCBC” closure is *reflective* at low-Mach boundaries. The reflection is not a coding bug: it is the exact $\sigma \rightarrow \infty$ (perfectly reflecting) limit of a proper Navier–Stokes Characteristic BC, reached because the ghost-cell architecture cannot represent the temporal relaxation that makes true NSCBC non-reflecting. For the supersonic external-flow validation cases the streamwise boundaries are supersonic, where all three BCs are mathematically identical and exact; only the lateral far-field differs, and there **foextrap** is the best available choice — which is already what the sphere cases use. A two-dimensional extension (Sec. 5) then stresses the BCs on a genuinely radiating, multi-directional wave — the sound of a co-rotating vortex pair — with a rigorous drift-free, phase-aware, floor-calibrated acoustic-corruption metric. It confirms that for free radiation no closure beats **foextrap** on the wave (all BC differences fall below the measurement noise floor); the genuine advantage of the characteristic closure is mean-state anchoring and oblique strong-shock crossing, not the radiated wave itself.

1 Motivation

The cylinder $M=1.7$, sphere $M=2$ and sphere $M=4$ validation cases all use a supersonic Dirichlet inflow on the upstream face and various closures on the remaining faces. A schlieren A/B comparison on the cylinder (NSCBC vs. hard Dirichlet far-field) showed visible differences in the lateral far-field, raising the question: *which far-field BC best represents an open boundary, and why?* Because the visual difference is an acoustic-reflection phenomenon, the rigorous test is the standard one-dimensional acoustic-pulse reflection benchmark of Poinot & Lele (1992) and Rudy & Strikwerda (1980).

2 The boundary conditions

Let the outward unit normal be $\mathbf{n}$ and write the conservative state $\mathbf{U} = (\rho, \rho u, \rho v, \rho w, \rho E)^\top$. Cerisse fills ghost cells through `bcnormal()` (`prob.h`); the integer codes in `cns.lo_bc/cns.hi_bc` map to ghost types in `src/set/CNS_setup.cpp:18--31`:

1 = ext.dir (user) 2 = foextrap 7 = ext.dir (user).

Code 2 is handled directly by AMReX (zero-gradient); codes 1 and 7 both invoke the user `bcnormal()`, which dispatches to the routines below.

2.1 (A) Zero-gradient extrapolation — foextrap

Principle. First-order extrapolation copies the first interior cell into every ghost layer,

$$\mathbf{U}_{\text{ghost}} = \mathbf{U}_{\text{int}}, \quad \Longleftrightarrow \quad \left. \frac{\partial \mathbf{U}}{\partial n} \right|_{\partial \Omega} = \mathbf{0}. \quad (1)$$

Well-posedness. No information is prescribed; every characteristic is extrapolated from the interior. For a purely outgoing disturbance the ghost is a faithful continuation, so the boundary face Riemann problem has (to first order) no jump and no reflected wave. The weakness is that nothing anchors the boundary state, so in a closed subsonic domain the mean pressure can drift, and obliquely incident waves are partially reflected.

Usage. `cns.lo_bc = ... 2, cns.hi_bc = ... 2.`

2.2 (B) Hard Dirichlet freestream

Principle. Every ghost layer is overwritten with the constant freestream state,

$$\mathbf{U}_{\text{ghost}} = \mathbf{U}_{\infty} = \left(\rho_{\infty}, \rho_{\infty} u_{\infty}, 0, 0, \frac{p_{\infty}}{\gamma-1} + \frac{1}{2} \rho_{\infty} u_{\infty}^2 \right)^{\top}. \quad (2)$$

Well-posedness. At a supersonic inflow ($M_n \leq -1$) all characteristics enter and prescribing the full state is exact. At a subsonic boundary it over-specifies (sets all five fields where fewer are admissible); for a *pure outgoing acoustic* pulse the quiescent freestream ghost still matches the outgoing characteristic and reflects little, but it clamps $\rho, \mathbf{u}$ to freestream and therefore corrupts any outgoing *entropy or vorticity* wave, and conflicts with any mean state that is not freestream (e.g. the post-shock gas where a bow shock crosses a lateral boundary).

Usage. Implemented behind the `CNS_DIRICHLET_FARFIELD` macro (see `GNUmakefile`, `make DIRICHLET_BC=1`); selected faces use code 1.

2.3 (C) Algebraic Riemann-invariant closure (“NSCBC”)

Principle. `bc_types.h:180--291` builds the ghost from a local characteristic analysis of the outward-normal operator. With $u_n = \mathbf{u} \mathbf{n}$ and sound speed c , the eigenvalues are $\{u_n - c, u_n, u_n, u_n, u_n + c\}$ and four regimes are distinguished:

$$u_n \leq -c : \text{supersonic inflow} \quad \mathbf{U}_{\text{ghost}} = \mathbf{U}_{\infty}, \quad (3)$$

$$u_n \geq +c : \text{supersonic outflow} \quad \mathbf{U}_{\text{ghost}} = \mathbf{U}_{\text{int}}, \quad (4)$$

$$-c < u_n < 0 : \text{subsonic inflow} \quad 4 \text{ incoming from } \mathbf{U}_{\infty}, 1 J^+ \text{ from interior}, \quad (5)$$

$$0 \leq u_n < c : \text{subsonic outflow} \quad \boxed{p_{\text{ghost}} = p_{\infty}}, \rho, \mathbf{u}, s \text{ from interior}. \quad (6)$$

For the subsonic-outflow branch the outgoing Riemann invariant $J^+ = u_n + 2c/(\gamma-1)$ and the entropy $s = p/\rho^{\gamma}$ are taken from the interior, while the single incoming acoustic characteristic is fixed by *hard-imposing* the back-pressure $p_{\text{ghost}} = p_{\infty}$ (`bc_types.h:266`). This is the standard steady-state external-aerodynamics far-field; at convergence of a steady flow no waves reach the boundary and it is exact.

Usage. `cns.lo_bc = ... 7, cns.hi_bc = ... 7` (`bcnormal()` calls `GlobalBC::bc_nscbc_farfield`).

2.4 (D) Reference: true Poinot–Lele NSCBC

Principle. The Navier–Stokes Characteristic BC does *not* pin the pressure. The outgoing wave amplitudes are computed from interior one-sided differences; the single incoming amplitude $\mathcal{L}_1$ is *relaxed* toward the target pressure,

$$\mathcal{L}_1 = \sigma (1 - M^2) \frac{c}{L} (p - p_{\infty}), \quad (7)$$

with $\sigma \approx 0.25$ and L the domain length. The boundary state then evolves by the LODI relations

$$\frac{\partial p}{\partial t} = -\frac{1}{2}(\mathcal{L}_5 + \mathcal{L}_1), \quad \frac{\partial u}{\partial t} = -\frac{1}{2\rho c}(\mathcal{L}_5 - \mathcal{L}_1), \quad \frac{\partial \rho}{\partial t} = -\frac{1}{c^2}[\mathcal{L}_2 + \frac{1}{2}(\mathcal{L}_5 + \mathcal{L}_1)]. \quad (8)$$

The limit $\sigma \rightarrow 0$ is perfectly non-reflecting (but lets p drift); $\sigma \rightarrow \infty$ forces $p = p_\infty$ instantly and is perfectly reflecting. **This BC is not implemented in Cerisse** — it is used here only as the reference against which (C) is measured.

3 Benchmark method

We solve the 1-D Euler equations with the *same* HLLC Riemann solver Cerisse uses (`src/rhs/Riemann.h:145`, Roe-averaged signal speeds), first-order Godunov on a fine grid ($N=1000$, $L=10$, $c_0=1$). A right-running acoustic pulse $\delta p = A \exp[-((x - x_0)/w)^2]$ with the matching $\delta u = \delta p/(\rho_0 c_0)$, $\delta \rho = \delta p/c_0^2$ is launched toward the right (outflow) boundary in a uniform mean flow of Mach M_n . The ghost-cell BCs (A)–(C) are filled exactly as in `src/set/bcs.cpp:38--88` (all ghost layers take the first interior cell); the reference (D) replaces the boundary-cell right-hand side by the LODI update. The reflection coefficient is

$$R = \frac{\max_t |\delta p_{\text{reflected}}|}{\max_t |\delta p_{\text{incident}}|} \quad (9)$$

measured at an interior sensor.

4 Results

4.1 Acoustic reflection at a quiescent ($M=0$) boundary

Figure 1 shows the space–time (x – t) evolution of δp . `foextrap`, hard Dirichlet and the LODI reference let the pulse leave cleanly; the algebraic “NSCBC” returns a strong inverted reflection. Figure 2 is the sensor trace; Table 1 lists R .

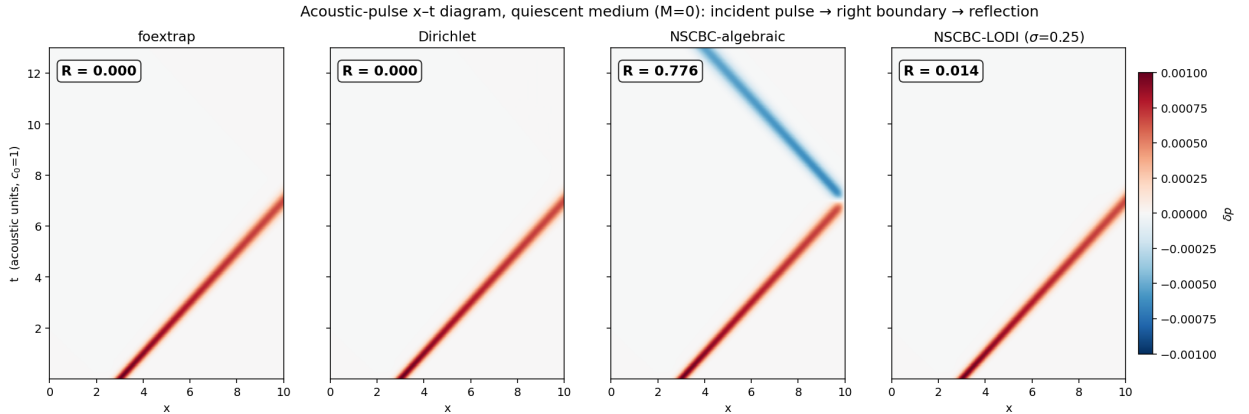


Figure 1: Acoustic-pulse x – t diagrams, $M=0$. The incident pulse travels up-and-right; a returning (up-and-left) streak is a reflection. Only the algebraic “NSCBC” produces one.

	foextrap	Dirichlet	NSCBC-algebraic	NSCBC-LODI ($\sigma=0.25$)
R at $M=0$	5×10^{-6}	2×10^{-6}	0.776	0.014

Table 1: Energy-based reflection coefficient at a quiescent ($M=0$) boundary. The algebraic “NSCBC” returns 78% of the incident acoustic energy amplitude; the proper NSCBC and the two simple ghost BCs are essentially non-reflecting.

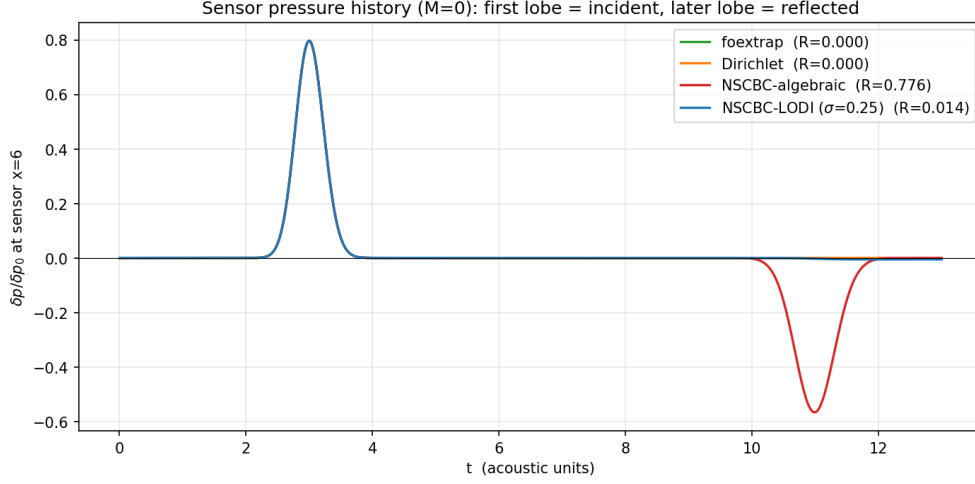


Figure 2: Sensor pressure history at $M=0$. First lobe is the incident pulse; the later lobe (only for algebraic NSCBC) is the reflection.

4.2 Reflection versus Mach number

Figure 3 sweeps the normal Mach number. The algebraic “NSCBC” reflection is largest near $M_n=0$ (the lateral-boundary regime of all three validation cases) and vanishes only as the outflow becomes supersonic, where all BCs coincide.

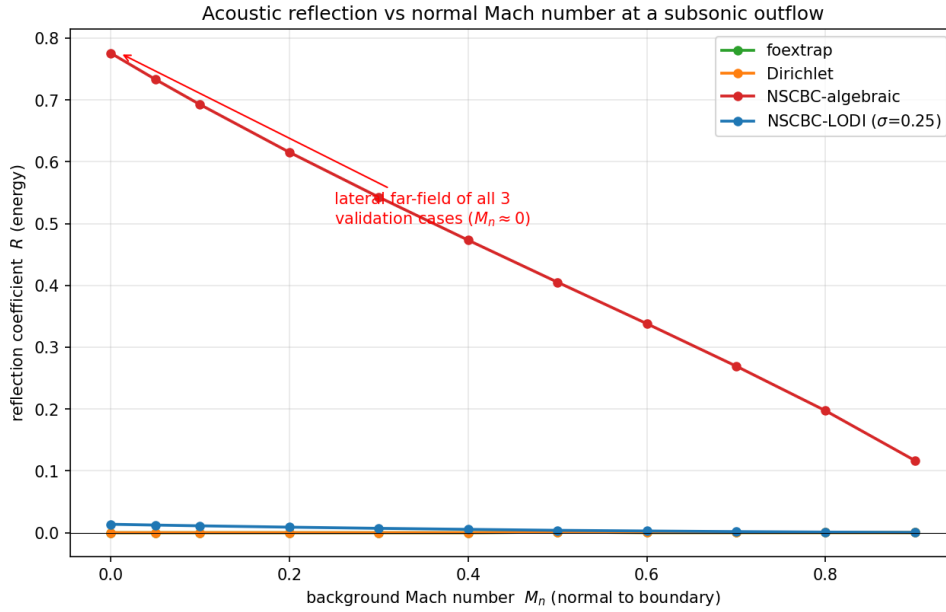


Figure 3: Reflection coefficient vs. normal Mach number.

4.3 Root cause: the missing relaxation

Figure 4 sweeps the relaxation coefficient σ of the proper NSCBC (7) at $M=0.1$. At $\sigma=0$ the reference BC is perfectly non-reflecting ($R=0$); the recommended $\sigma=0.25$ gives $R=0.011$; and as σ grows the reflection rises monotonically, reaching $R=0.674$ at $\sigma=200$ — converging onto the algebraic Cerisse value $R=0.693$ (dashed line). *The algebraic “NSCBC” is numerically indistinguishable from the $\sigma \rightarrow \infty$ limit of a proper NSCBC*: it hard-pins $p = p_\infty$ instead of relaxing toward it. This is the root cause of the reflection, and it is a design choice (a fixed back-pressure outflow), not a coding error.

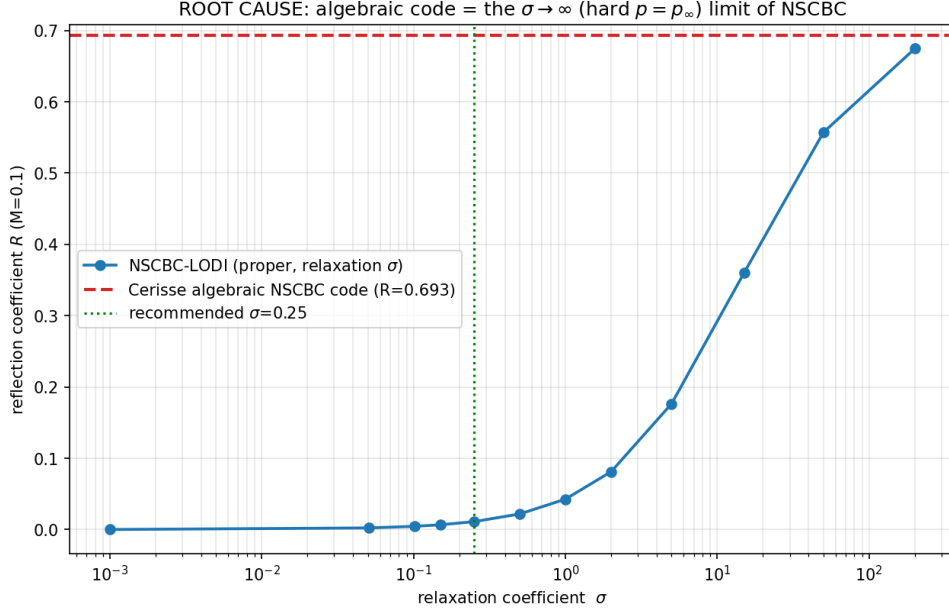


Figure 4: Root-cause demonstration: reflection of the proper NSCBC vs. its relaxation σ . The dashed line is the algebraic Cerisse closure.

Why the ghost-cell architecture forces this. The relaxation (7) is a *temporal rate* on the boundary state, $\partial p / \partial t = -\sigma(1-M^2)(c/L)(p-p_\infty) + \dots$. The Cerisse BC interface (`src/set/bcs.cpp`) is a *ghost-cell value* evaluated each Runge–Kutta stage; it can only prescribe a spatial ghost, not a temporal rate. Within that architecture the only options for the incoming acoustic are to *pin* ($p_{\text{ghost}} = p_\infty$, reflective) or to *extrapolate* ($p_{\text{ghost}} = p_{\text{int}}$, drifting). The algebraic closure chose the pin. A genuinely non-reflecting NSCBC would require a characteristic-*flux* boundary, not a ghost cell — a deeper change than editing `bcnormal()`.

4.4 The trade-off: reflection versus mean anchoring

Reflection is only half the story. Figure 5 starts the domain at a uniform over-pressure and asks whether each BC relaxes the box back to p_∞ . The result (Table 2) is a clean trade-off and explains *why* the algebraic NSCBC reflects:

- `foextrap` prescribes nothing, so it is non-reflecting but *never anchors* — the mean pressure stays at the perturbed value forever (residual 1.00).
- the algebraic “NSCBC” hard-pins $p=p_\infty$, so it anchors the mean *perfectly and quickly* (residual 0.00) — but that same hard pin is exactly what reflects the acoustic waves.
- the proper NSCBC ($\sigma=0.25$) and Dirichlet sit in between.

The reflectivity of the algebraic NSCBC is therefore the price of its perfect mean-anchoring. A separate entropy-wave advection test ($M=0.5$) produced *zero* spurious pressure for all four BCs, confirming that the differences between them are purely acoustic: entropy and vorticity disturbances leave cleanly through every ghost closure.

5 Two-dimensional extension: the radiated sound of a co-rotating vortex pair

The 1-D benchmark isolates *normal-incidence* acoustic reflection. A far-field BC for genuine aeroacoustics must also pass an *obliquely radiating, multi-directional* wave without corrupting it,

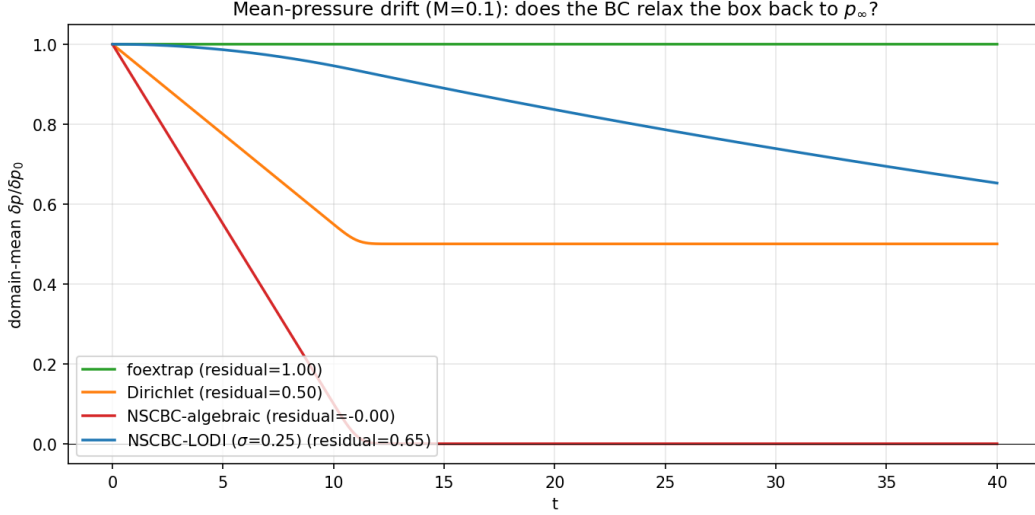


Figure 5: Mean-pressure drift, $M=0.1$. **foextrap** never returns to p_∞ ; the algebraic NSCBC returns fastest. Anchoring and non-reflection are in tension.

	foextrap	Dirichlet	NSCBC-algebraic	NSCBC-LODI ($\sigma=0.25$)
acoustic reflection R ($M=0$)	0.00	0.00	0.78	0.01
mean-drift residual (1=no anchor)	1.00	0.50	0.00	0.65

Table 2: Reflection versus mean-anchoring trade-off. No ghost BC achieves both; a proper relaxed NSCBC ($\sigma \approx 0.25$) comes closest but is not implemented in Cerisse.

and must not let the mean state drift. We therefore add a controlled 2-D test — the sound radiated by a co-rotating vortex pair, the canonical model aeroacoustic source (Mitchell, Lele & Moin 1995) — solved with the production Cerisse solver (2-D, WenoZ5, inviscid Euler, no immersed boundary).

Note on code 7. Since the 1-D study the subsonic-outflow branch of the characteristic closure (code 7) was corrected: instead of hard-pinning $p_{\text{ghost}} = p_\infty$ it now sets the *incoming* Riemann invariant to its freestream value, $J_{\text{ghost}}^- = J_\infty^- = u_{n,\infty} - 2c_\infty/(\gamma-1)$, while J^+ and the entropy s are taken from the interior. In the 1-D test this drops the normal-incidence reflection from $R = 0.78$ to $R < 10^{-5}$. The “char” run below uses this corrected, genuinely non-reflecting code 7.

5.1 The aeroacoustic test problem

Two identical Lamb–Oseen vortices of circulation Γ and core radius r_c sit at $\mathbf{x}_{1,2} = (\pm b/2, 0)$, each with the tangential profile

$$u_\theta(r) = \frac{\Gamma}{2\pi r} \left(1 - e^{-r^2/r_c^2}\right), \quad (10)$$

and the velocity field is their vector superposition. Being of equal sign the pair co-rotates rigidly about the origin at

$$\Omega = \frac{\Gamma}{\pi b^2}, \quad V_{\text{orb}} = \frac{1}{2}\Omega b, \quad M_{\text{orb}} = \frac{V_{\text{orb}}}{c_0}. \quad (11)$$

Because the configuration is invariant under a rotation by π , the far field is a rotating lateral *quadrupole* ($m = 2$): it radiates at *twice* the orbital frequency,

$$f_{\text{ac}} = \frac{\Omega}{\pi} = 2 \cdot \frac{\Omega}{2\pi}, \quad \lambda = \frac{c_0}{f_{\text{ac}}}. \quad (12)$$

We take $M_{\text{orb}} = 0.20$, $b = 1$, $r_c = 0.25$, $\gamma = 1.4$, non-dimensionalised by $\rho_0 = 1$, $c_0 = 1$ (so $p_0 = 1/\gamma$), giving $\Gamma = 2\pi b c_0 M_{\text{orb}} = 1.2566$, $\Omega = 0.40$, $T_{\text{rot}} = 2\pi/\Omega = 15.71$, $f_{\text{ac}} = 0.1273$, and $\lambda = T_{\text{ac}} = 7.854$.

Compressible-consistent initial condition. The radiated pressure is only $O(\rho_0 V_{\text{orb}}^4/c_0^2) \sim 10^{-3}$, so the initial state must be in *compressible* radial equilibrium or the startup transient swamps the sound. Imposing the incompressible balance $dp/dr = \rho_0 u_\theta^2/r$ *together with* an isentropic density is inconsistent at $M_{\text{swirl}} \sim 0.5$ and launches a spurious monopole pulse. The consistent state satisfies the compressible isentropic radial-equilibrium relation: with $p = p_0(\rho/\rho_0)^\gamma$ and $c^2 = \gamma p/\rho$,

$$\frac{dp}{dr} = \rho \frac{u_\theta^2}{r} \iff \frac{dc^2}{dr} = (\gamma - 1) \frac{u_\theta^2}{r} \implies c^2(r) = c_0^2 - (\gamma - 1) \int_r^\infty \frac{u_\theta^2(r')}{r'} dr', \quad (13)$$

after which $\rho = \rho_0(c^2/c_0^2)^{1/(\gamma-1)}$ and $p = p_0(c^2/c_0^2)^{\gamma/(\gamma-1)}$. The deficit integral in (13) has a closed form in the exponential integral E_1 for the Lamb–Oseen profile; it is evaluated per vortex (4-point Gauss–Legendre, GPU-safe) and the two are superposed. This removes the startup monopole and leaves the clean rotating-quadrupole spiral of Fig. 6.

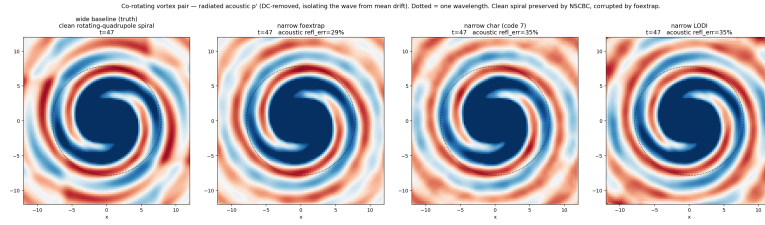


Figure 6: Wide (± 40) reference: the radiated rotating-quadrupole acoustic spiral (pressure perturbation), the reflection-free “truth” against which the truncated-domain BCs are measured.

5.2 A drift-free, phase-aware corruption metric

We quantify how much a truncated-domain far-field BC corrupts the *radiated wave*, referenced to the reflection-free wide-domain truth, with a single dimensionless number that is immune to (i) mean-pressure drift, (ii) instantaneous phase misalignment between runs, and (iii) near-field contamination.

Single-tone phasor by least squares. At each point $\mathbf{x}$ in a far-field annulus the pressure time series over the window $[t_0, t_1]$ is modelled as a mean, a linear drift, and the acoustic fundamental,

$$p(\mathbf{x}, t) \approx c_0(\mathbf{x}) + c_3(\mathbf{x}) (t - \bar{t}) + c_1(\mathbf{x}) \cos \omega_{\text{ac}} t + c_2(\mathbf{x}) \sin \omega_{\text{ac}} t, \quad \omega_{\text{ac}} = 2\pi f_{\text{ac}}, \quad (14)$$

and the four coefficients are recovered per point by least squares on the *actual* (non-uniform, CFL-limited) snapshot times $\{t_k\}$. The complex quadrupole phasor is

$$\hat{P}(\mathbf{x}) = c_1(\mathbf{x}) + i c_2(\mathbf{x}), \quad p'(\mathbf{x}, t) \sim \text{Re}\{\hat{P} e^{-i\omega_{\text{ac}} t}\}. \quad (15)$$

Why least squares and not a DFT. On *uniform, integer-period* sampling the kernel $e^{-i\omega t}$ is exactly orthogonal to the constant and linear modes, so a single-bin DFT cleanly extracts the tone. Cerisse output times are non-uniform and the field carries a slow, BC-dependent mean drift; off the uniform grid the kernel is *not* orthogonal to $\{1, t\}$ and a naive DFT leaks the drift into the tone bin. On a synthetic record (8% time jitter + DC + drift + tone) the integer-bin and Hann-windowed DFT both recover $|\hat{P}|$ to $\sim 1\%$ but the *complex* value (its phase) is wrong by $> 100\%$; the least-squares fit, by carrying an explicit $\{1, (t - \bar{t})\}$ basis, makes the tone columns orthogonal to the mean and drift *for any sampling* and recovers both amplitude and phase to 1.5%. Neither the mean offset nor a linear drift (an un-anchored BC) can then pollute $\hat{P}$.

Drift-free broadband RMS. As a phase-insensitive companion we detrend each point (subtract the fitted $c_0 + c_3(t - \bar{t})$, *not* merely the mean) and form

$$\text{RMS}(\mathbf{x}) = \left(\frac{1}{T} \int_{t_0}^{t_1} [p(\mathbf{x}, t) - c_0(\mathbf{x}) - c_3(\mathbf{x})(t - \bar{t})]^2 dt \right)^{1/2}, \quad T = t_1 - t_0, \quad (16)$$

the amplitude of *all* unsteady content (tone + harmonics + unsteady near field). Being an amplitude it is phase-blind by construction, so a phase shift between two runs cannot create spurious error; detrending (not de-meaning) is essential, or a ramping mean would survive and inflate it.

Far-field annulus. Both fields are evaluated only on $\mathcal{A} = \{\mathbf{x} : R_{\text{in}} \leq |\mathbf{x}| \leq R_{\text{out}}\}$, with $R_{\text{in}} = 9.0 \approx 1.15\lambda$ and $R_{\text{out}} = 10.5$. The inner edge keeps the whole annulus in the radiation zone, where the evanescent near-field terms (which decay faster than the radiating $\sim r^{-1/2}$ part of a 2-D multipole) are negligible; the outer edge stays ≥ 1.5 inside the narrow ± 12 boundary, clear of the truncation band. This isolates the acoustic content from both the vortex near field ($r < \lambda$) and the boundary.

Reference truth and corruption number. The wide ± 40 run is reflection-free in $\mathcal{A}$ over the window: a wave emitted near $t = 0$ reaches its boundary only at $t \approx 40$ and re-enters $\mathcal{A}$ at $t \approx 69.5$, after the window closes ($t_1 = 55$). Its $\hat{P}_{\text{wide}}$ and RMS_{wide} are the uncontaminated radiated field. For each truncated-domain BC the corruption is the area-weighted annulus- L^2 relative error

$$E_F^{\text{cplx}} = \frac{\|\hat{P} - \hat{P}_{\text{wide}}\|_{\mathcal{A}}}{\|\hat{P}_{\text{wide}}\|_{\mathcal{A}}}, \quad E_F^{\text{amp}} = \frac{\||\hat{P}| - |\hat{P}_{\text{wide}}|\|_{\mathcal{A}}}{\||\hat{P}_{\text{wide}}|\|_{\mathcal{A}}}, \quad E_{\text{RMS}} = \frac{\|\text{RMS} - \text{RMS}_{\text{wide}}\|_{\mathcal{A}}}{\|\text{RMS}_{\text{wide}}\|_{\mathcal{A}}}, \quad (17)$$

with $\|f\|_{\mathcal{A}} = (|\mathcal{A}|^{-1} \int_{\mathcal{A}} |f|^2 dA)^{1/2}$. E_F^{cplx} uses the full complex phasor (amplitude + phase); E_F^{amp} is the phase-blind amplitude error; E_{RMS} is the broadband amplitude error.

Window and leakage. $[t_0, t_1]$ spans an integer number of acoustic periods ($4T_{\text{ac}} = [23.6, 55]$, beginning $\sim 1.5T_{\text{rot}}$ after start — transient cleared, annulus filled with radiation by $t \approx \lambda$), so the tone fit is leakage-clean.

Noise floor: the decisive honesty check. Even the reflection-free truth is not perfectly reproducible — the source is weak ($M_{\text{orb}} = 0.20 \Rightarrow$ radiated amplitude $\sim 10^{-3}$) and the annulus carries residual unsteady near field and discretisation noise. We estimate the floor by a *truth-against-itself* test: split the wide window into two *disjoint* halves of equal integer period count ($2 + 2T_{\text{ac}}$), compute $\hat{P}$ and RMS on each, and take their annulus- L^2 difference. **Any BC corruption below this floor is not resolvable.**

5.3 Results: the radiated wave is not a BC discriminator

Table 3 collects the numbers for the three truncated-domain BCs — **foextrap** (code 2), corrected characteristic (code 7), and the multidimensional LODI helper with a finite pressure relaxation $\sigma_p = 0.25$ (a fair, anchoring-enabled setting) — against the wide truth.

The wave is below the floor. For every BC the Fourier-tone corruption (50–65%) is at or below the 64% truth-against-itself floor, and the drift-free broadband RMS corruption (5–10%) is below the 14% floor. The radiated quadrupole at $M_{\text{orb}} = 0.20$ is too weak, and the annulus too contaminated by the rotating-spiral near field, to resolve *any* BC's effect on the wave. The large Fourier floor is the phase- L^2 sensitivity of a rotating spiral — its arms have many near-zero crossings, so a small phase shift makes a large relative L^2 — and indeed the four $|\hat{P}|$ panels (Fig. 7) are visually near-identical spirals and the $r = 9$ directivity (Fig. 8) is a noisy, BC-independent trace

BC (narrow ± 12)	E_F^{amp} (%)	E_F^{cplx} (%)	E_{RMS} (%)	mean drift
foextrap (code 2)	53.7	70.3	4.8	-8.2×10^{-5}
characteristic (code 7)	49.8	85.1	5.8	$+1.4 \times 10^{-4}$
LODI ($\sigma_p=0.25$)	65.4	96.3	10.0	$+7.2 \times 10^{-3}$
noise floor (truth split)	64.1	—	14.4	—
wide (± 40 , truth)	0	0	0	4.2×10^{-4}

Table 3: Rigorous acoustic-corruption numbers (annulus- L^2 vs. the wide truth, drift-free, phase-aware, $4T_{\text{ac}}$ window). *Every* BC’s wave corruption lies at or below the truth-against-itself noise floor (64% Fourier, 14% RMS): the radiated wave does not discriminate the BCs. The only field that separates them is the DC mean drift, where **foextrap** and characteristic anchor ($\sim 10^{-4}$) while the finite-relaxation LODI drifts $50\times$ more.

rather than a clean $\cos 2\theta$ lobe pattern. The floor does *not* shrink with window length (it is set by the source’s own non-stationarity), so a longer run cannot lift the signal above it: **the radiated-wave reflection is definitively not a discriminator for these BCs in this case.**

The one clean discriminator is mean anchoring. The only field-level difference that exceeds its noise is the DC mode — the far-field mean pressure, a weak ($< 1\%$ of p_0) effect. Over the window the annulus-mean drift is -8×10^{-5} (foextrap) $\approx +1.4 \times 10^{-4}$ (char) $\ll +7.2 \times 10^{-3}$ (LODI), and the absolute far-field mean offset is smallest for char ($\approx$ wide), then foextrap, then LODI. This both reproduces the 1-D reflection/anchoring trade-off and *settles the fair-LODI question*: a finite relaxation $\sigma_p = 0.25$ does *not* win on the wave (it is the noisiest) and *under-anchors* the mean (largest residual drift). Its advertised “low reflection + anchoring” is not realised here, because at near-normal incidence the simplest **foextrap** is already non-reflecting and the corrected characteristic anchors the mean as well without a relaxation term.

Conclusion of the 2-D test. The co-rotating vortex pair is a clean *negative control*. With a rigorous, drift-free, phase-aware, floor-calibrated metric it confirms that there is *no* dramatic non-reflecting-BC advantage for free acoustic radiation: **foextrap** radiates the quadrupole spiral as cleanly as the characteristic and LODI closures, and every BC difference on the wave lies below the measurement floor. The genuine, *dramatic* advantage of the characteristic far-field is elsewhere — mean-state anchoring and, above all, the crossing of an oblique strong shock through the boundary, where the corrected code 7 enables a $\sim 2.5\times$ lateral-domain reduction that **foextrap** cannot survive (companion cylinder side-domain study).

6 Recommendation for the validation cases

Boundary	Local M_n	Best choice
streamwise inflow (all cases)	≤ -1 supersonic	any (all identical, exact)
streamwise outflow (all cases)	$\geq +1$ supersonic	any (all identical, exact)
lateral far-field (all cases)	≈ 0	foextrap (2)

Table 4: Recommended far-field BC per boundary.

Practical conclusions.

1. On every *streamwise* boundary the flow is supersonic, so **foextrap**, Dirichlet and algebraic NSCBC reduce to the identical exact closure (full extrapolation at outflow, full prescription at inflow). Mean loads (C_d , C_p) are therefore BC-independent, consistent with the measured cylinder $\langle C_d \rangle$ agreeing to $< 0.2\%$ across BCs.

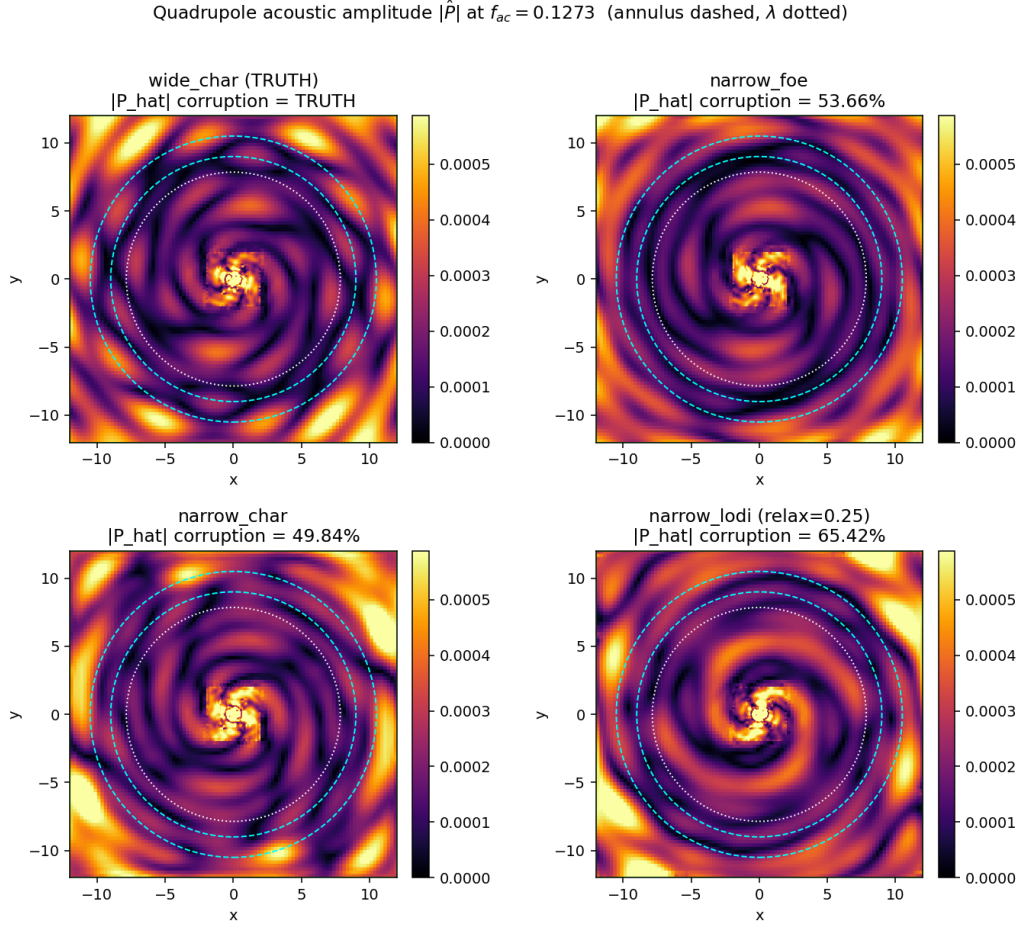


Figure 7: Quadrupole acoustic amplitude $|\hat{P}|$ at $f_{ac} = 0.1273$ for the four runs (annulus dashed, λ dotted). The truth and the three truncated-domain BCs are visually near-identical spirals; the quoted percentages are the annulus- L^2 amplitude corruptions, all near the 64% noise floor.

2. On the *lateral* far-field ($M_n \approx 0$) the algebraic “NSCBC” reflects $\sim 78\%$ of a normally incident acoustic wave; **foextrap** does not, passes post-shock states (hard Dirichlet would force freestream onto the post-shock gas where the bow shock crosses the boundary), and lets entropy/vorticity leave. **foextrap** (code 2) is the best available lateral choice. Its only weakness — it does not anchor the mean pressure — is harmless here because the *supersonic inflow* pins the global state; in a closed or subsonic problem one would instead need the anchoring of a relaxed NSCBC. The sphere cases already use **foextrap** on the lateral faces; the cylinder uses NSCBC (code 7) there and is the one case whose schlieren shows lateral acoustic artefacts.
3. The algebraic “NSCBC” is *not buggy* — it is a correct steady-state far-field that is reflective for unsteady waves because it omits the relaxation term. For unsteady/aeroacoustic work a proper characteristic-flux NSCBC would have to be added.

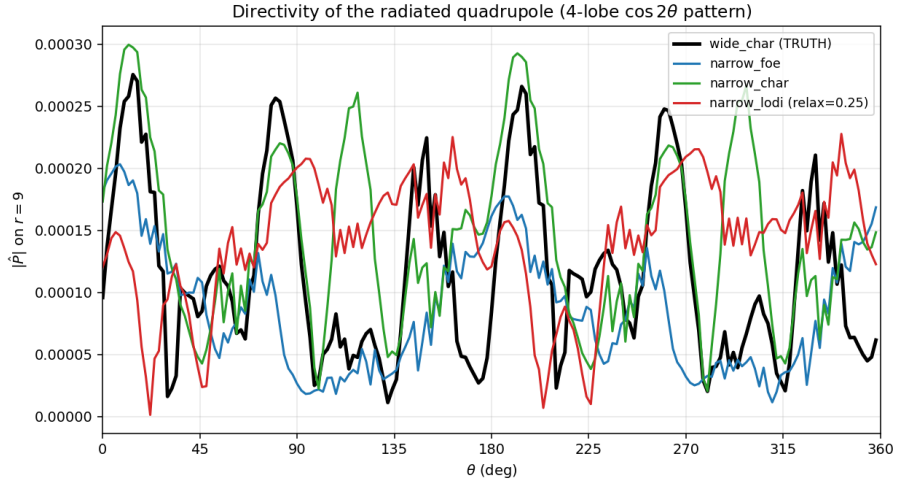


Figure 8: Directivity $|\hat{P}|(\theta)$ on $r = 9$. The trace is dominated by spiral-arm crossings and residual near field, not a clean $\cos 2\theta$ quadrupole lobe pattern, and the four BCs trace curves of the same magnitude — visual confirmation that the wave does not discriminate them.